

Plotting First Order Solution Families (rachel batra)

Plotting the family of solutions of a first order differential equation : $y' = x^2$ where $y(0)=k$

a) Using the pre defined function : ode2 (works for an ode of order upto 2)

```
eqn:'diff(y,x)=x^2;
sol:ode2(eqn,y,x);
sol1:ic1(sol,x=0,y=k);
v1:ev(sol1,k=-2);
v2:ev(sol1,k=-1);
v3:ev(sol1,k=1);
v4:ev(sol1,k=2);
wxplot2d([rhs(v1),rhs(v2),rhs(v3),rhs(v4)], [x,-3,3],[style,[lines,1],[lines,2],[lines,3],[lines,4]])$

-- eqn : ' diff ( y , x ) = x ^ 2 ;                               /* The equation is: y'= x^2 */
> sol : ode2 ( eqn , y , x ) ;                                     /*sol is the name given to the general solution */
sol1 : ic1 ( sol , x = 0 , y = k ) ;                               /*sol1 is a particular solution, with default
constant %c being replaced by */
v1 : ev ( sol1 , k = - 2 ) ;                                       /*random values are given to k in the next set of
statements */
v2 : ev ( sol1 , k = - 1 ) ;
v3 : ev ( sol1 , k = 1 ) ;
v4 : ev ( sol1 , k = 2 ) ;
wxplot2d ( [ rhs ( v1 ) , rhs ( v2 ) , rhs ( v3 ) , rhs ( v4 ) ] , [ x , - 3 , 3 ] , [ style , [ lines , 1 ] , [ lines , 2
] , [ lines , 3 ] , [ lines , 4 ] ] ) $
```

$$\frac{d}{dx}y = x^2$$

$$y = \frac{x^3}{3} + \%c$$

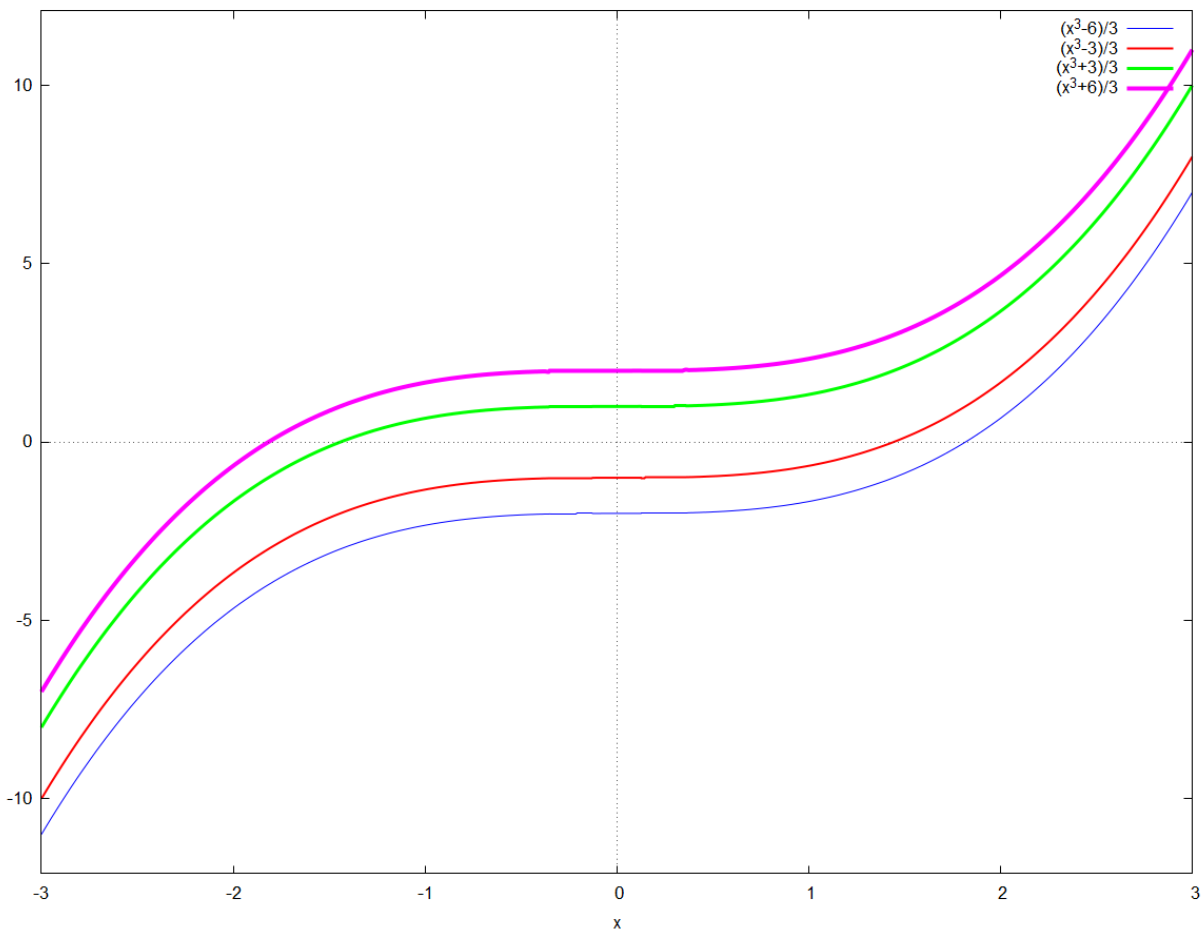
$$y = \frac{x^3 + 3k}{3}$$

$$y = \frac{x^3 - 6}{3}$$

$$y = \frac{x^3 - 3}{3}$$

$$y = \frac{x^3 + 3}{3}$$

$$y = \frac{x^3 + 6}{3}$$



b) Using the pre-defined function: `desolve` (works for an ode of any order)

```
-- kill ( all ) $ /*to unassign the previously defined variables */
> eqn : diff ( y ( x ) , x ) = x ^ 2 ; /* y is explicitly written as a function of x.
Also, no single quote required. */
sol : desolve ( eqn , y ( x ) ) ; /*desolve doesnt give constants explicitly but
their value(s0) */
sol1 : ev ( sol , y ( 0 ) = k ) ;
v1 : ev ( sol1 , k = - 2 ) ;
v2 : ev ( sol1 , k = - 1 ) ;
v3 : ev ( sol1 , k = 1 ) ;
v4 : ev ( sol1 , k = 2 ) ;
wxplot2d ( [ rhs ( v1 ) , rhs ( v2 ) , rhs ( v3 ) , rhs ( v4 ) ] , [ x , - 3 , 3 ] , [ style , [ lines , 1 ] , [ lines , 2
] , [ lines , 3 ] , [ lines , 4 ] ] ) $
```

$$\frac{d}{dx}y(x) = x^2$$

$$y(x) = \frac{x^3}{3} + y(0)$$

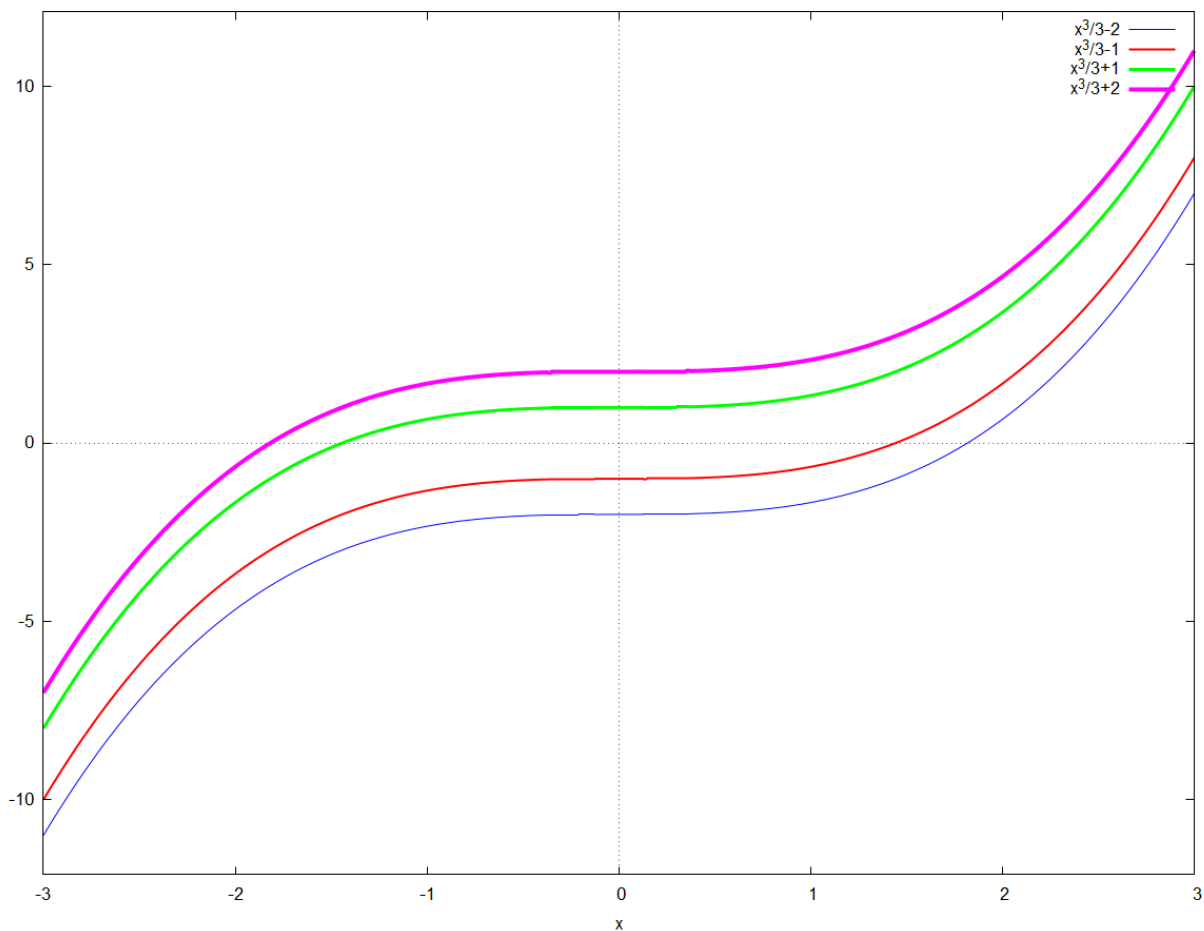
$$y(x) = \frac{x^3}{3} + k$$

$$y(x) = \frac{x^3}{3} - 2$$

$$y(x) = \frac{x^3}{3} - 1$$

$$y(x) = \frac{x^3}{3} + 1$$

$$y(x) = \frac{x^3}{3} + 2$$



```
-- kill ( all ) $
> eqn : ' diff ( y , x ) = 9 . 8 - 0 . 196 · y ;
sol : ode2 ( eqn , y , x ) ;
sol1 : ic1 ( sol , x = 0 , y = k ) ;
constant %c being replaced by */
v1 : ev ( sol1 , k = - 2 ) ;
statements */
v2 : ev ( sol1 , k = - 1 ) ;
v3 : ev ( sol1 , k = 1 ) ;
v4 : ev ( sol1 , k = 2 ) ;
wxplot2d ( [ rhs ( v1 ) , rhs ( v2 ) , rhs ( v3 ) , rhs ( v4 ) ] , [ x , - 3 , 3 ] , [ style , [ lines , 1 ] , [ lines , 2
] , [ lines , 3 ] , [ lines , 4 ] ] ) $
```

$$\frac{d}{dx}y = 9.8 - 0.196y$$

%default: replaced -9.8 by -49/5 = -9.8

%default: replaced 0.196 by 49/250 = 0.196

%default: replaced -9.8 by -49/5 = -9.8

%default: replaced 0.196 by 49/250 = 0.196

%default rat: replaced -9.8 by $-49/5 = -9.8$

%default rat: replaced 0.196 by $49/250 = 0.196$

$$y = e^{-\frac{49x}{250}} \left(50e^{\frac{49x}{250}} + c \right)$$

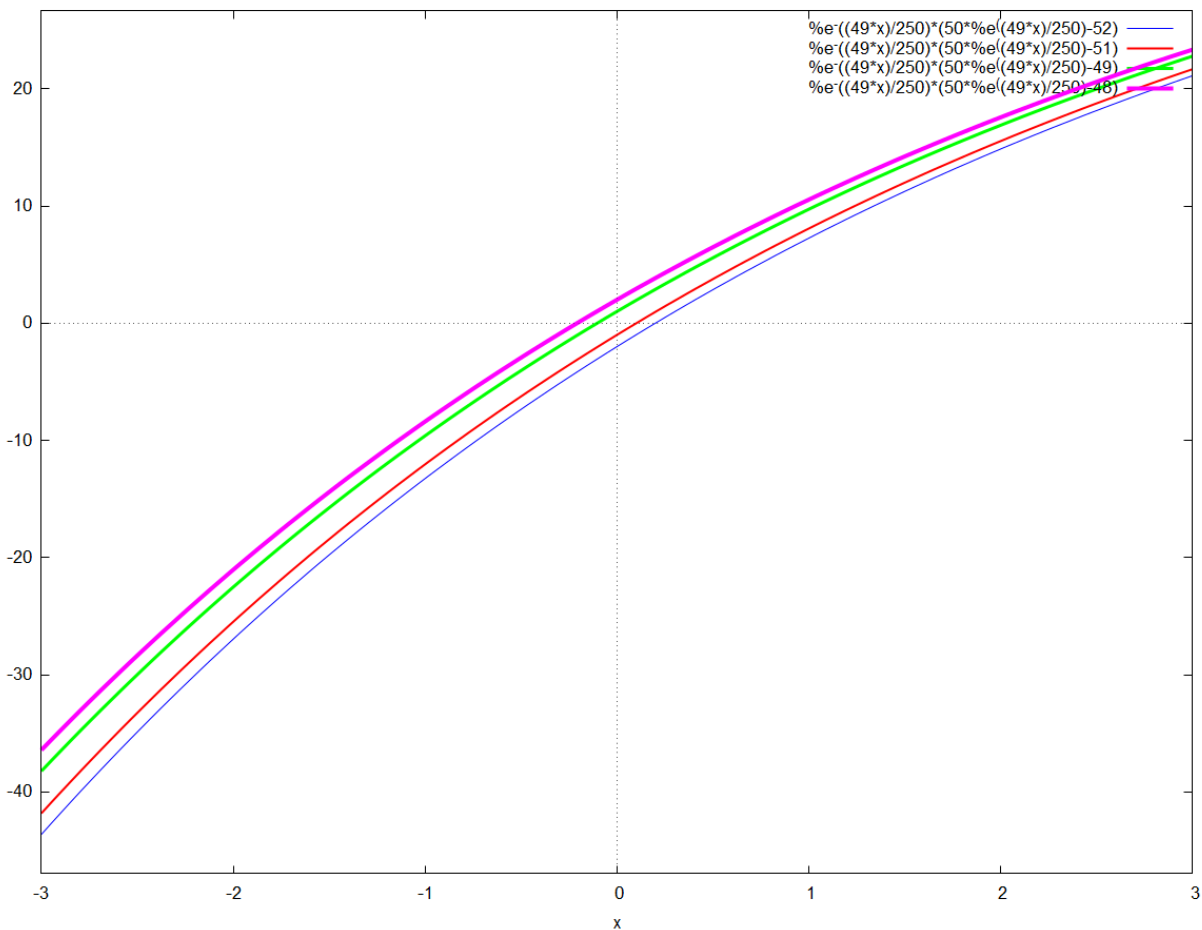
$$y = e^{-\frac{49x}{250}} \left(50e^{\frac{49x}{250}} + k - 50 \right)$$

$$y = e^{-\frac{49x}{250}} \left(50e^{\frac{49x}{250}} - 52 \right)$$

$$y = e^{-\frac{49x}{250}} \left(50e^{\frac{49x}{250}} - 51 \right)$$

$$y = e^{-\frac{49x}{250}} \left(50e^{\frac{49x}{250}} - 49 \right)$$

$$y = e^{-\frac{49x}{250}} \left(50e^{\frac{49x}{250}} - 48 \right)$$



```
-- kill ( all ) $
> eqn : ' diff ( y , x ) · cos ( x ) + y · sin ( x ) = 2 · ( cos ( x ) ) ^ 3 · ( sin ( x ) ) - 1 ;
sol : ode2 ( eqn , y , x ) ;
sol1 : ic1 ( sol , x = 0 , y = k ) ;
v1 : ev ( sol1 , k = 17 ) ;
v2 : ev ( sol1 , k = - 8 ) ;
v3 : ev ( sol1 , k = 0 ) ;
v4 : ev ( sol1 , k = 125 ) ;
wxplot2d ( [ rhs ( v1 ) , rhs ( v2 ) , rhs ( v3 ) , rhs ( v4 ) ] , [ x , - 5 , 5 ] , [ style , [ lines , 2 ] , [ lines , 8
] , [ lines , 4 ] , [ lines , 7 ] ] ) $
```

$$\cos(x) \left(\frac{d}{dx} y \right) + \sin(x) y = 2 \cos(x)^3 \sin(x) - 1$$

$$y = \cos(x) \left(-\frac{1}{\tan(x)^2 + 1} - \tan(x) + \%c \right)$$

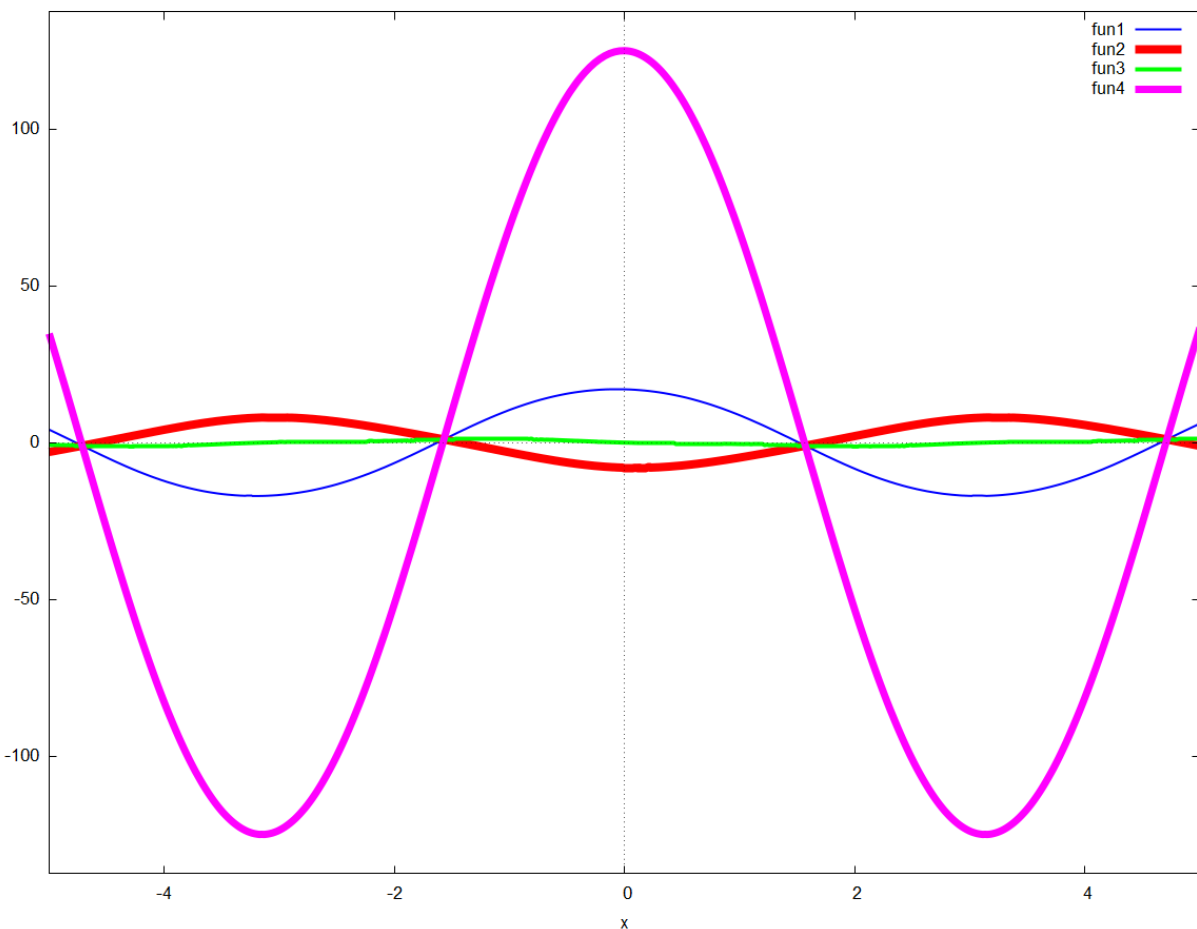
$$y = -\frac{\cos(x)\tan(x)^3 + (-k-1)\cos(x)\tan(x)^2 + \cos(x)\tan(x) - k\cos(x)}{\tan(x)^2 + 1}$$

$$y = -\frac{\cos(x)\tan(x)^3 - 18\cos(x)\tan(x)^2 + \cos(x)\tan(x) - 17\cos(x)}{\tan(x)^2 + 1}$$

$$y = -\frac{\cos(x)\tan(x)^3 + 7\cos(x)\tan(x)^2 + \cos(x)\tan(x) + 8\cos(x)}{\tan(x)^2 + 1}$$

$$y = -\frac{\cos(x)\tan(x)^3 - \cos(x)\tan(x)^2 + \cos(x)\tan(x)}{\tan(x)^2 + 1}$$

$$y = -\frac{\cos(x)\tan(x)^3 - 126\cos(x)\tan(x)^2 + \cos(x)\tan(x) - 125\cos(x)}{\tan(x)^2 + 1}$$



```
-- eqn : 'diff ( y , x ) - y = sin ( x ) + cos ( x ) ;
> sol : ode2 ( eqn , y , x ) ;
sol1 : ic1 ( sol , x = 0 , y = k ) ;
v1 : ev ( sol1 , k = - 1 ) ;
v2 : ev ( sol1 , k = - 3 ) ;
v3 : ev ( sol1 , k = 0 ) ;
```

```

v3 : ev ( sol1 , k = 3 );
wxplot2d ( [ rhs ( v1 ) , rhs ( v2 ) , rhs ( v3 ) , rhs ( v4 ) ] , [ x , - 5 , 5 ] , [ style , [ lines , 1 ] , [ lines , 2
] , [ lines , 3 ] , [ lines , 4 ] ] );

```

$$\frac{d}{dx}y - y = \sin(x) + \cos(x)$$

$$y = \%e^x \left(\frac{\%e^{-x} (\sin(x) - \cos(x))}{2} + \frac{\%e^{-x} (-\sin(x) - \cos(x))}{2} + \%c \right)$$

$$y = (k + 1)\%e^x - \cos(x)$$

$$y = -\cos(x)$$

$$y = -\cos(x) - 2\%e^x$$

$$y = \%e^x - \cos(x)$$

$$y = 4\%e^x - \cos(x)$$

